



JOURNAL OF QUALITY ASSURANCE IN AGRICULTURAL SCIENCES (JQAAS)

Comparative Incidence of Thrips tabaci on Bt and Non-Bt Cotton Varieties under Field Conditions

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Article Received 12-10-2025, Article Revised 20-5-2026, Article Accepted 20-07-2026

ABSTRACT

Thrips tabaci is the main interruption of the cotton production in last ten years among the sucking insect pests of cotton in Sindh Pakistan. In the present study, five varieties of cotton (Koong, IR-3701, MNH-886, Naib-878 and non-Bt CRIS-585, CRIS-613, Saami and Nusrat) were evaluated for their resistance against thrips tabaci under field condition at Tandojam, Sindh, Pakistan during the crop season 2024. Thrips tabaci population was appeared on all Bt and non-Bt cotton varieties after the germination of cotton crop. The maximum population of T. tabaci was recorded on the Bt cotton variety Naib-878 (22.978 ± 1.464 /plant), while the minimum population of t. tabaci was found on the non-Bt cotton variety Nusrat (8.443 ± 0.884 /plant), followed by Bt varieties Koong (18.998 ± 1.096 /plant), IR-3701 (15.780 ± 1.017 /plant), MNH-886 (13.233 ± 0.918 /plant) and non Bt CRIS-585 (12.765 ± 0.983 /plant), CRIS-613 (11.768 ± 0.958 /plant), and Saami (10.202 ± 0.915 /plant), respectively. No significant differences were found between the Bt and non-Bt cotton varieties (MNH-886, CRIS-585, CRIS-613, Saami) and t. tabaci. There was some variability in the yield per acre of Bt and Non-Bt cotton types. Bt cotton varieties showed high yield potential with MNH-886 yielding highest (35.133 per acre) followed by IR-3701 (18.511 per acre) among the varieties. The varieties Koong and Naib-878 had 14.778 and 12.778 per acre, respectively, with comparatively low productivity. Among the varieties, Non-Bt cotton varieties, in which Nusrat grew with the highest yield (27.156 per acre), was found to be ranked second overall. Variety Saami yielded 12.311 per acre of crop. Lower yields were observed in CRIS-613 (10.178 per acre) and CRIS-585 (8.956 per acre). Overall, maximum temperature had the greatest effect on the all varieties of cotton (Bt and non-Bt), followed by minimum temperature and humidity. Wind speed moderately and significantly affected both types of cotton with Bt cotton slightly more sensitive than the non-Bt cotton. It finds that after the germination of the cotton crop, there were significant differences in the population of Thrips tabaci among most Bt and non-Bt cotton varieties. The Bt variety Naib-878 had the highest t. tabaci population while the non-Bt variety Nusrat had the lowest infestation level. No significant differences in t. tabaci population was observed among the varieties MNH-886, CRIS-585, CRIS-613 and Saami suggesting comparable susceptibility among these varieties.

Keywords: Bt and Non- Bt cotton varieties, Thrips tabaci, resistant, weather, yield

INTRODUCTION

Cotton (*Gossypium hirsutum* L.) has been a basic natural resource since the beginning of various civilizations and is still one of the important plant species for human needs (Shahrajabian, et al., 2020). The cotton considered to support the textile industry and contributes about 55% of the total edible oil production Pakistan (Shah et al., 1999; Zardari, et al., 2022). The various factors involving in reducing the cotton production day by day such as, climate change, insect pest resistant and disease problems (Mohyuddin et al., 1997; Muunaweera

et al., 2022). The insect pests are responsible for yield losses of cotton crop about 20-40% in Pakistan (Ramzan, et al., 2019). There are about 166 insects reported on cotton crops (Rajendran, et al., 2018) from which some insect pests are responsible for significant damage to cotton crop are classified as sucking and chewing types (Blaise and Kranthi, 2019). The sucking insect *thrips tabaci*, *amrasca bigutula bigutula*, *phenacoccus solenopsis* and chewing types *spodoptra litura*, *earias* sp., *halicoverpa armigera*, *pactinophora gossypiella* and *oxycarneous laetus* are most destructive pests for cotton

crop (Arshad et al., 2011; Ashfaq et al., 2011; Shah, 2014; Nasir, et al., 2021). Among the sucking insect pest, the *thrips tabaci* has been reported a main risk for reducing the cotton production in Pakistan ((Kumar, et al., 2014; Asif, et al., 2022). It is polyphagous pest and found in the field throughout the year due availability of alternate host plants (Asif, et al., 2020). Now days *Thrips tabaci* causes heavy damage on *Bt* and non-*Bt* cotton genotypes in comparison of other sucking insect pests such as thrips, whitefly and jassid (Masood, et al., 2019). It has been smaller in size and body in cylinder-shaped, both the nymph and the adult sucking cell sap from under surface of the cotton leaves (Arif, et al., 2006; Salman, et al., 2011; Ben-Yakir, et al., 2020; David, 2020). The damage caused by *thrips tabaci*, the silvery appearance on the leaves, which reduces the photosynthesis process of cotton plant leaves and decrease the production (Hashmi, 1994; Arif et al., 2006; Khan et al., 2008). The Losses of *thrips tabaci* during severe infestations ranged from 10-40% of the cotton yield (Attique and Ahmad, 1990; Gahukar, 2006). The use of chemical on large scale in agriculture for pest control is generally considered the cheapest and most effective method of pest control. However, improper use of pesticides creates various problems for living organisms (Malik et al., 2015; Boudh, and Singh, 2019), the injurious impact of pesticides on the environment, natural biodiversity, pollinators, food chains, and health outcomes (Ali et al., 2021). The indiscriminate use of insecticides kills the natural enemies, develop pest resurgen, pollute the environment and direct causes losses to human (Salman et al., 2011; Ramzan, 2019), and non-target organism about 98% are directly or indirectly affected by the synthetic chemicals. Several studies have estimated that 80% of pesticide sprays directly pollute the environment. The host plant resistance is the most important practice for the management of insect pest, and it can control the insect population (Khan et al., 2010). Cultivation of genotypes with resistant traits is the most operative, environmentally friendly, and economical method that is most successful in increasing crop production (Khan et al., 2011). The present study conducted to screening the *Bt* and non- *Bt* cotton varieties against *thrips tabaci* under field condition of Tandojam, Sindh Pakistan.

MATERIALS AND METHODS

Site and Location: The experiment was carried out in the field of Plant Protection Research Institute, Agriculture Research Sindh, Tandojam Pakistan, during cotton cropping season 2024. Field selection was done keeping in mind appropriate pest pressure, soil fertility status, climatic condition, and availability of optimum farm inputs in the study area.

Cotton varieties: Total eight cotton varieties were used in this study. The four *Bt* and four non-*Bt* cotton varieties were selected to monitor the *thrips tabaci* population. The four *Bt* varieties such as Koong, IR-3701, MNH-886,

Naib-878 and four non-*Bt* CRIS-585, CRIS-613, Sami, and Nusrat were obtained from Cotton Research Institute, Tandojam, Sindh.

Experimental layout: The experiment was arranged randomized complete block design (RCBD) with three replications. The recommended seed rate of cotton 2 to 3 kg/acre were used, moreover plant to plant distance 30 cm and row to row distance 75 cm respectively. All agronomic practices were carried out as per the recommendation.

Procedure for counting pest population: The data were taken on weekly basis from germination up to picking of cotton. There were no any pesticides was application use against sucking insect pests of cotton crop. The five randomly selected plants from each treatment were observed upper, middle, and the lower leaves of cotton plant for counting of *thrips tabaci* nymph and adult population per treatment

Meteorological data: Weather such as Maximum Temperature, Minimum Temperature, Humidity and wind velocity were observed.

Statistical analysis: The data was used to analyze the analysis of variance (ANOVA) the (LSD) 5% probability of the least significant difference. The Correlation were achieved to determine the impact of various weather parameters on the population fluctuation of *T. tabaci* on various *Bt* and non *Bt* cotton varieties. The analyzes of data were performed using STATISTIX 8.1 computer software.

RESULTS

The results of present study showed that the highly significant difference ($F=19.82$, $P<0.05$) was recorded on the population of *thrips tabaci* on various *Bt* varieties of cotton. According to result (Table 1.) indicated that the maximum weekly mean population of *thrips tabaci* was recorded in the first week of observation on Naib-878 (8.178 ± 2.063 /plant) and the lowest population was found on MNH-886 (5.356 ± 1.511 /plant) followed by Koong 7.556 ± 1.855 /plant and IR-3701 6.089 ± 1.513 /plant respectively. Moreover, that the *t. tabaci* population increased slowly and reached peak population. The peak population of *t. tabaci* was found on eleventh week of observation. The highest mean population of *thrips tabaci* was recorded on Naib-878 was (30.289 ± 5.385 /plant) and lowest population was observed on MNH-886 (18.378 ± 4.273 /plant) followed by Koong (18.378 ± 4.273 /plant) and IR-3701 (20.533 ± 4.442 /plant) respectively. After that the population of *t. tabaci* move decreased to wards was rcorred in last week of observation. The minimum population of *t. tabaci* (9.178 ± 3.23 /plants) was recorded on MNH-886, while the maximum population (23.044 ± 4.64 /plant) was found Naib-878, followed by Konnj (17.756 ± 4.133 /plant) and IR-3701 (10.800 ± 3.41 /plant) respectively.

Table. 1 Weekly mean population of *thrips tabaci* on different *Bt* cotton varieties

Week	Koong	IR-3701	MNH-886	Naib-878
22-04-2021	7.556±1.855 ^{s-w}	6.089±1.513 ^{vw}	5.356±1.511 ^w	8.178±2.063 ^{r-w}
29-04-2021	8.356±1.743 ^{p-w}	7.267±1.683 ^{t-w}	6.311±1.730 ^{uvw}	9.556±1.999 ^{n-m}
06-05-2021	15.644±3.461 ^{e-w}	11.978±2.859 ^{k-w}	8.289±2.539 ^{q-w}	17.733±3.849 ^{b-t}
13-05-2021	17.133±3.413 ^{e-v}	15.378±3.910 ^{f-w}	11.844±3.159 ^{l-w}	19.489±3.949 ^{a-o}
20-05-2021	18.178±3.662 ^{b-t}	16.178±3.640 ^{e-w}	12.711±3.311 ^{j-w}	20.778±3.721 ^{a-m}
27-05-2021	19.444±4.190 ^{a-p}	17.089±3.735 ^{c-v}	13.511±3.524 ^{h-w}	21.844±4.482 ^{a-m}
03-06-2021	20.156±3.913 ^{a-o}	18.067±3.480 ^{b-t}	14.822±3.476 ^{g-w}	22.089±4.505 ^{a-l}
10-06-2021	22.400±4.909 ^{a-l}	19.822±4.226 ^{a-o}	15.844±4.058 ^{e-w}	24.178±4.814 ^{a-h}
17-06-2021	22.356±4.747 ^{a-l}	20.267±4.499 ^{a-o}	16.222±3.941 ^{e-w}	26.378±4.756 ^{a-f}
24-06-2021	23.089±4.161 ^{a-k}	19.289±4.242 ^{a-r}	17.267 4.032 ^{c-u}	28.067±4.478 ^{a-c}
01-07-2021	24.111±4.410 ^{a-i}	20.533±4.442 ^{a-n}	18.378±4.273 ^{b-t}	30.289±5.385 ^a
08-07-2021	23.378±4.889 ^{a-j}	19.378±4.280 ^{a-q}	17.867±4.277 ^{b-t}	29.822±4.980 ^a
15-07-2021	22.111±4.463 ^{a-l}	18.422±4.118 ^{b-s}	16.622±4.437 ^{d-v}	28.467±5.313 ^{ab}
22-07-2021	21.778±4.581 ^{a-m}	17.556±3.976 ^{b-t}	15.822±3.921 ^{e-w}	27.489±4.993 ^{a-d}
29-07-2021	20.489±4.757 ^{a-n}	16.956±4.150 ^{c-v}	13.867±3.584 ^{h-w}	26.733±5.073 ^{a-e}
05-08-2021	19.533±4.674 ^{a-o}	15.956±3.929 ^{e-w}	12.889±3.493 ^{j-w}	25.156±4.700 ^{a-g}
12-08-2021	18.489±4.182 ^{b-s}	13.022±3.709 ^{i-w}	11.400±3.385 ^{l-w}	24.311±4.727 ^{a-h}
19-08-2021	17.756±4.133 ^{b-t}	10.800±3.412 ^{m-w}	9.178±3.231 ^{o-w}	23.044±4.648 ^{a-k}

*Means followed by the same letters are not significantly different (LSD 2.6221, P < 0.05)

Similarly, the *t. tabaci* was recorded on non-*Bt* cotton varieties. The results of study showed significant difference in all non-*Bt* cotton varieties. The results of study showed significant (F=7.28, P<0.05) difference was recorded on different non-*Bt* varieties of cotton. The data indicated that the (Table 2) initial maximum population of *t. tabaci* on non-*Bt* varieties of cotton such as CRIS-585 (4.311±1.373/plant), CRIS-613 (3.044±1.162/plant), Saami (2.244±0.726/plant) while the minimum *t. tabaci* population was found Nusrat (1.156±0.353/plant) cotton variety. The population of *t. tabaci* increased gradually on all non *bt* cotton varieties

and reached peak population. The highest *t. tabaci* Population on non-bt varieties on CRIS-585 (19.311±4.145/plant) and minimum population on Nusrat (14.556±2.937/plant) followed by CRIS-613 (17.756±3.784/plant) and Saami (16.156±3.276/plant) respectively. After that the *t. tabaci* population decreased slowly on non-*Bt* cotton varieties. The maximum population of *t. tabaci* (9.556±2.73/plant) on CRIS-613, while the minimum population (6.311±2.17/plant) on Nusrat non *Bt* followed by Saami (7.911±2.48/plant) and CRIS-585 (8.444±2.65/plant) on last week of observation respectively.

Table. 2 Weekly mean population of *thrips tabaci* on different non- *Bt* cotton varieties

Weeks	CRIS-585	CRIS-613	Saami	Nusrat
22-04-2021	4.311±1.373 ^{l-r}	3.044±1.162 ^{o-r}	2.244±0.726 ^{qr}	1.156±0.353 ^r
29-04-2021	5.244±1.583 ^{j-r}	4.156±1.435 ^{m-r}	3.844±1.144 ^{n-r}	2.556±0.874 ^{qr}
06-05-2021	8.667±2.396 ^{c-r}	6.956±1.911 ^{g-r}	5.822±1.710 ^{i-r}	3.889±1.300 ^{n-r}
13-05-2021	10.889±3.215 ^{b-o}	9.778±2.686 ^{b-q}	7.333±2.011 ^{c-r}	4.889±1.718 ^{k-r}
20-05-2021	11.733±3.024 ^{a-n}	10.378±2.724 ^{b-q}	8.711±2.302 ^{c-r}	5.689±1.588 ^{i-r}
27-05-2021	12.156±3.097 ^{a-m}	11.600±2.898 ^{a-n}	9.778±2.715 ^{b-q}	7.333±2.250 ^{f-r}
03-06-2021	13.644±3.538 ^{a-i}	12.533±3.236 ^{a-l}	10.933±2.367 ^{b-o}	8.978±2.348 ^{c-r}
10-06-2021	14.556±3.080 ^{a-g}	13.156±3.508 ^{a-j}	11.311±2.607 ^{a-n}	10.822±2.487 ^{b-o}
17-06-2021	15.844±4.087 ^{a-e}	14.489±3.839 ^{a-h}	13.711±3.904 ^{a-i}	11.667±3.298 ^{a-n}
24-06-2021	16.822±3.790 ^{abc}	15.467±3.682 ^{a-f}	14.556±3.450 ^{a-g}	12.578±2.824 ^{a-k}
01-07-2021	19.311±4.145 ^a	17.756±3.784 ^{ab}	16.156±3.276 ^{a-e}	14.556±2.937 ^{a-g}
08-07-2021	17.667±3.986 ^{ab}	16.644±4.185 ^{a-d}	15.489±3.819 ^{a-f}	12.489±3.663 ^{a-l}
15-07-2021	16.533±3.784 ^{a-d}	15.511±3.616 ^{a-f}	13.644±3.562 ^{a-i}	11.311±2.889 ^{a-n}
22-07-2021	15.756±3.763 ^{a-e}	14.933±3.852 ^{a-g}	12.756±3.488 ^{a-k}	10.956±3.081 ^{b-o}
29-07-2021	14.889±3.601 ^{a-g}	13.711±3.180 ^{a-i}	10.756±2.470 ^{b-p}	9.978±2.634 ^{b-q}
05-08-2021	12.178±3.504 ^{a-m}	11.733±3.343 ^{a-n}	9.956±2.710 ^{b-q}	8.978±2.056 ^{c-r}
12-08-2021	11.133±3.328 ^{a-n}	10.422±3.527 ^{b-q}	8.733±2.607 ^{c-r}	7.844±2.312 ^{c-r}
19-08-2021	8.444±2.656 ^{c-r}	9.556±2.730 ^{b-q}	7.911±2.484 ^{c-r}	6.311±2.173 ^{h-r}

*Means followed by the same letters are not significantly different (LSD1.9422, P < 0.05)

The results of overall mean population of *thrips tabaci* of various *Bt* and non-*Bt* varieties of cotton (Figure. 1). The significant difference in population of *t. tabaci* was recorded on all *Bt* and non- *Bt* cotton varieties after the germination of crops. Overall maximum mean population of *thrips tabaci* was recorded on *Bt* variety Naib 878 (22.978 ± 1.464 /plant) and lowest population of *t. tabaci* was found (8.443 ± 0.884 /plant) on non *Bt* cotton

variety Nusrat followed by Koong (18.998 ± 1.096 /plant) IR-3701(15.780 ± 1.017 /plant), MNH-886 (13.233 ± 0.918 /plant), CRIS-585 (12.765 ± 0.983 /plant), CRIS-613 (11.768 ± 0.958 /plant) and Saami (10.202 ± 0.915 /plant) respectively. There is no significant difference was recorded on MNH-886, CRIS-585, CRIS-613 and Saami respectively

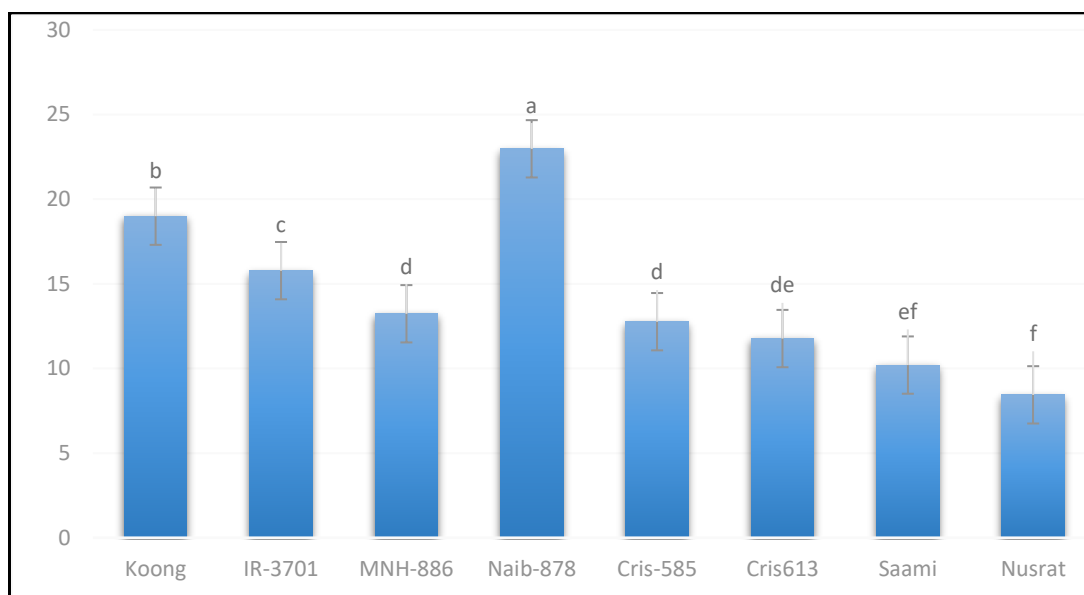


Figure 1. Over all mean population of *Thrips tabaci* on *Bt* and Non-*Bt* varieties of cotton

Correlation of weather Parameters with *Bt* and Non-*Bt* Cotton Varieties: The data indicate (Table 3.) that the relationship between weather parameters and cotton performance differed slightly between *Bt* and non-*Bt* cotton varieties, though most parameters showed statistically significant correlations ($P < 0.005$). For *Bt* cotton, maximum temperature exhibited a strong positive correlation ($r = 0.8789$; $P < 0.005$), indicating that increases in maximum temperature were strongly associated with improved performance. Minimum temperature showed a weak but significant positive correlation ($r = 0.2114$; $P < 0.005$). Relative humidity

also had a weak positive association ($r = 0.2329$; $P < 0.005$), while wind speed demonstrated a moderate positive correlation ($r = 0.3694$; $P < 0.005$). Similarly, in non-*Bt* cotton, maximum temperature was strongly and positively correlated with crop performance ($r = 0.8468$; $P < 0.005$). Minimum temperature again showed a weak but significant positive relationship ($r = 0.2128$; $P < 0.005$). Humidity exhibited a slightly stronger positive correlation compared to *Bt* cotton ($r = 0.3090$; $P < 0.005$), and wind speed showed a moderate positive association ($r = 0.3271$; $P < 0.005$).

Table 3. Impact of weather on population *t. tabaci* on *Bt* and non *Bt* cotton varieties

Weather Parameter	<i>Bt</i> cotton	Non- <i>Bt</i> cotton
Maximum Temperature	$r=0.878869497$; $P < 0.005$	$r=0.846773705$; $P < 0.005$
Minimum Temperature	$r=0.211387958$; $P < 0.005$	$r=0.212756611$; $P < 0.005$
Humidity	$r=0.232911243$; $P < 0.005$	$r=0.309032569$; $P < 0.005$
wind	$r=0.369394687$; $P < 0.005$	$r=0.327058109$; $P < 0.005$

Effect of *t. tabaci* population on yield performance of *Bt* and Non-*Bt* Cotton Varieties

The data indicated (Table 4.) that the yield per acre of *Bt* and Non-*Bt* cotton varieties showed noticeable variation. Among the *Bt* cotton varieties, MNH-886 produced the highest yield (35.133 per acre), indicating superior yield potential. This was followed by IR-3701 with a yield of

18.511 per acre. The varieties Koong and Naib-878 yielded 14.778 and 12.778 per acre, respectively, showing comparatively lower productivity. In the Non-*Bt* cotton varieties, Nusrat recorded the highest yield (27.156 per acre), ranking second overall among all varieties. The variety Saami produced a moderate yield of 12.311 per acre. Lower yields were observed in CRIS-

613 (10.178 per acre) and CRIS-585 (8.956 per acre). Overall, *Bt* cotton varieties generally exhibited higher yield performance than Non-*Bt* cotton varieties, mainly due to the outstanding yield of MNH-886. However, the

Non-*Bt* variety Nusrat also showed strong yield potential compared to several *Bt* varieties, highlighting the role of varietal differences in yield performance.

Table 4. Effect of *t. tabaci* population on yield performance of *Bt* and Non-*Bt* Cotton Varieties

Bt cotton varieties	Varieties	Yield/acre
	IR-3701	18.511
	Koong	14.778
	Naib-878	12.778
	MNH-886	35.133
Non-Bt cotton varieties	Saami	12.311
	CRIS-613	10.178
	Nusrat	27.156
	CRIS-585	8.956

DISCUSSION

According to the results, higher population of *thrips tabaci* was recorded on *Bt* varieties compared to non-*Bt* varieties. The findings of present study showed that the maximum population of *T. tabaci* was recorded on the *Bt* cotton variety Naib-878 (22.978 ± 1.464 /plant), while the minimum population of *t. tabaci* was found on the non-*Bt* cotton variety Nusrat (8.443 ± 0.884 /plant), followed by *Bt* varieties Koong (18.998 ± 1.096 /plant), IR-3701 (15.780 ± 1.017 /plant), MNH-886 (13.233 ± 0.918 /plant) and non *Bt* CRIS-585 (12.765 ± 0.983 /plant), CRIS-613 (11.768 ± 0.958 /plant), and Saami (10.202 ± 0.915 /plant), respectively. Therefore, our findings are supported by Naveen et al., (2007) and Main et al. (2005) who reported that higher *thrips tabaci* populations were recorded on *Bt* cotton than conventional cotton. While according to Whitehouse et al. (2005) reported that the *thrips tabaci* populations were lower on *Bt* cotton compared to conventional cotton cultivars while most studies showed that sucking insect pests were found in equal abundance in *Bt* and non-*Bt* cultivars. Sisterson et al. (2004) mentioned that the results of the present study show higher population of *thrips tabaci* on *Bt* cotton varieties which may be due to different response and favorable environmental conditions. The results of current study indicated that the correlation analysis revealed that all studied weather parameters had a positive and statistically significant relationship with both *Bt* and Non-*Bt* cotton, indicating that climatic factors play a crucial role in influencing cotton performance. However, the strength of these relationships varied among parameters and between cotton types, highlighting differential sensitivity to environmental conditions. The results of Naveen et al. (2007) and Men et al. (2005). Population growth or decline can be caused by temperature and relative humidity which further affect the intrinsic rate of natural growth of *thrips tabaci* (Morai, 2000) while according to Li et al. (1992), various climatic factors such as temperature, relative humidity were found to be positively correlated with *thrips tabaci* population on cotton. Wahla et al., (1996) who found that rainfall and

relative humidity and minimum air temperature were negatively correlated with *thrips tabaci* population. According to Ananthakrishnan (1993), Kirk (1997) and Ligotowska (1997), climatic variables including rainfall, temperature, relative humidity and wind have been reported as important factors that significantly affect *thrips tabaci* populations. Relatively high temperature and lack of rainfall are associated with increased onion thrips *tabaci* population, while high relative humidity and rainfall reduce the thrips *tabaci* population (Hamdi and Salem, 1994). Sharma and Pampati (2006) in a study found no significant difference in whitefly and thrips populations on *Bt* and non-*Bt* cotton. During the finding of this study, it was observed that the highest population of *thrips tabaci* was recorded on both *Bt* and non *Bt* varieties of cotton in 1st week July 2021 and then decline to wards maturity. These finding accordance with the work of Nizamain et al., (2002) reported that the population of *thrips tabaci* rapidly increase in the starting of July and then slowly decreased in the month of August. Khan et al., (2012) reported that the peak population of *thrips tabaci* was recorded in month of July. Similarly, Nahiyoan et al. (2016) supported that the peak population of *thrips tabaci* was recorded from the month of July and August. Therefore, finding of current study the *Bt* varieties showed high population of *thrips tabaci* while the non-*Bt* varieties found less attack of *Thrips tabaci*. The overall outcome of this study confirms that maximum population of *thrips tabaci* was recorded on *Bt* as compared with non-*Bt* varieties of cotton respectively. The yield performance of *Bt* and Non-*Bt* cotton varieties exhibited considerable variation, reflecting inherent genetic differences among varieties and their response to prevailing environmental conditions. Overall, *Bt* cotton varieties showed superior yield performance compared to Non-*Bt* varieties, although certain Non-*Bt* varieties demonstrated competitive productivity. Attique and Ahmed (1990) supported our findings that the attack by *thrips tabaci* (14.6 per leaf) and jasside (4.6 per leaf) caused a 37.6% losses in yield of cotton seed. An other study mentioned by Kerns et al., (2019) reported that the density of *thrips tabaci* and feeding injury from *thrips*

tabaci direct impact of growth of seedling and reducing the yield of cotton.

CONCLUSION

It is concludes that the population of *Thrips tabaci* varied significantly among most *Bt* and non-*Bt* cotton varieties after crop germination. The *Bt* variety Naib-878 exhibited the highest *t. tabaci* population, whereas the non-*Bt* variety Nusrat showed the lowest infestation level. However, no significant differences in *t. tabaci* population were observed among the varieties MNH-886, CRIS-585, CRIS-613, and Saami, indicating similar levels of susceptibility among these varieties.

CONFLICT OF INTEREST

Author have declared no conflict of interest.

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